

Student Course Registration System — Schema v1.0

students		
PK	student_id	INT AUTO_INCREMENT
UK	student_number	VARCHAR(20) UNIQUE NOT NULL
	first_name	VARCHAR(50) NOT NULL
	last_name	VARCHAR(50) NOT NULL
UK	email	VARCHAR(100) UNIQUE NOT NULL
	password_hash	VARCHAR(255) NOT NULL (bcrypt)
	created_at	TIMESTAMP DEFAULT NOW()

courses		
PK	course_id	INT AUTO_INCREMENT
UK	course_code	VARCHAR(20) UNIQUE NOT NULL
	course_name	VARCHAR(100) NOT NULL
	instructor	VARCHAR(100) NOT NULL
	max_capacity	INT NOT NULL DEFAULT 40
	current_enrollment	INT NOT NULL DEFAULT 0
	semester	VARCHAR(20) NOT NULL
	created_at	TIMESTAMP DEFAULT NOW()

admins		
PK	admin_id	INT AUTO_INCREMENT
UK	username	VARCHAR(50) UNIQUE NOT NULL
	password_hash	VARCHAR(255) NOT NULL (bcrypt)
	created_at	TIMESTAMP DEFAULT NOW()

registrations		
PK	registration_id	INT AUTO_INCREMENT
FK	student_id	INT NOT NULL → students
FK	course_id	INT NOT NULL → courses
	registered_at	TIMESTAMP DEFAULT NOW()
	status	ENUM 'ACTIVE' 'DROPPED'
UK	UNIQUE (student_id, course_id)	

LEGEND	
PK	Primary Key — uniquely identifies each row
FK	Foreign Key — links to another table
UK	Unique Constraint — no duplicate values
—	Solid line — One-to-Many relationship (1:N)
- - -	Dashed line — Out-of-scope reference

CARD 3 — CHECKLIST	
✓	Define tables (students, courses, registrations, admins)
✓	Map primary / foreign keys + unique constraints
✓	Generate .sql schema file (schema_v1.sql)